

Practical – 10

[Disk Scheduling]

Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

INPUT :

```
GNU nano 8.7 sstf.c
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int request[8] = {98, 183, 41, 122, 14, 124, 65, 67};
    int visited[8] = {0};
    int head = 53;
    int total = 0;
    int i, j, min, index;

    for(i = 0; i < 8; i++)
    {
        min = 1000;
        for(j = 0; j < 8; j++)
        {
            if(visited[j] == 0)
            {
                int dist = abs(head - request[j]);
                if(dist < min)
                {
                    min = dist;
                    index = j;
                }
            }
        }

        total = total + min;
        head = request[index];
        visited[index] = 1;

        printf("Move to cylinder %d\n", head);
    }

    printf("\nTotal head movement = %d", total);

    return 0;
}
```

OUTPUT :

```
HP@Hiteshri MSYS ~
$ nano sstf.c

HP@Hiteshri MSYS ~
$ gcc sstf.c -o sstf

HP@Hiteshri MSYS ~
$ ./sstf
Move to cylinder 41
Move to cylinder 65
Move to cylinder 67
Move to cylinder 98
Move to cylinder 122
Move to cylinder 124
Move to cylinder 183
Move to cylinder 14
Total head movement = 323
```